

PYTHON DATA ANALYTICS

FINAL PROJECT REPORT

Technical Support Dataset

2,330 Records • 22 Columns

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Dataset URL

<https://www.kaggle.com/datasets/suvroo/technical-support-dataset>

Project Objective

The objective of this project was to analyse technical support ticket data to understand ticket characteristics, response and resolution performance, SLA compliance, support channels, ticket topics, agent-group performance, customer satisfaction, and monthly ticket trends. The analysis covered data loading, preprocessing, exploratory data analysis, statistical summaries, and visualizations.

1. Dataset Overview

The Technical Support Dataset contains 2,330 support-ticket records and 22 columns. The variables include ticket status, ticket ID, priority, source, topic, agent group, agent name, created time, expected SLA timestamps, response and resolution times, SLA status, close time, agent interactions, survey results, product group, support level, country, latitude, and longitude. The dataset contains a mixture of categorical, numerical, date-time, and duration-related variables. This variety made it suitable for demonstrating data cleaning, transformation, exploratory analysis, statistical analysis, and visualization.

2. Data Pre-processing

2.1 Missing Values

Missing values were identified in First response time (18), Resolution time (418), Close time (1,157), Agent interactions (18), and Survey results (1,157). Missing values were handled during preprocessing. Mode imputation was used for the time-related columns, while median imputation was used for the numerical columns Agent interactions and Survey results. After imputation, the dataset contained no missing values.

2.2 Duplicate Records

The dataset was checked for duplicate rows. The duplicate check returned 0 duplicate records. Therefore, no rows were removed, and all 2,330 records were retained.

2.3 Correcting Data Types

Date-related columns were converted to datetime format. Duration columns were converted to timedelta format and then represented in minutes for analysis and visualization.

2.4 Derived Columns

Four derived columns were created: First response time (minutes), Resolution time (minutes), Created Date, and Created Month. These columns supported response-time analysis, resolution analysis, and monthly ticket trends.

2.5 Filtering and Aggregation

The data was filtered to identify high-priority tickets, resulting in 416 high-priority tickets. Aggregation was performed to calculate average resolution time by priority and agent group.

3. Exploratory Data Analysis

3.1 Univariate Analysis

Status analysis showed 1,173 Closed tickets, 739 Resolved tickets, 400 In Progress tickets, and 18 Open tickets. Priority analysis showed 1,192 Low, 722 Medium, and 416 High-priority tickets. Email was the most common source with 1,234 tickets, followed by Chat with 850 and Phone with 246. Topic analysis showed Product setup as the most common topic with 630 tickets, followed by Pricing and licensing with 457 and Feature request with 417. The dataset also contains both “Pricing and licensing” and “Pricing and Licensing” as separate categories because of capitalization differences.

3.2 Bivariate Analysis

Average first response time was approximately 29.64 minutes for Chat, 29.83 minutes for Email, and 28.74 minutes for Phone. By priority, average first response time was approximately 29.91 minutes for High, 29.22 minutes for Low, and 30.19 minutes for Medium. Average resolution time by priority was approximately 24.71 minutes for High, 23.48 minutes for Low, and 25.44 minutes for Medium.

3.3 Multivariate Analysis

When Priority, Source, and First response time were analysed together, response times varied across combinations. For High-priority tickets, Phone had the lowest average response time at approximately 27.20 minutes and Email the highest at approximately 30.81 minutes. For Low priority, Phone was lowest at approximately 28.21 minutes and Chat highest at approximately 29.74 minutes. For Medium priority, Chat was lowest at approximately 29.70 minutes and Phone highest at approximately 30.62 minutes.

3.4 Groupby and Pivot Analysis

Average resolution time was approximately 24.58 minutes for 1st line support and 23.44 minutes for 2nd line support. The pivot table showed different resolution-time averages across priority and source combinations. For example, High-priority Phone tickets averaged approximately 25.68 minutes, while High-priority Email tickets averaged approximately 24.39 minutes.

3.5 Correlation Analysis

Most numerical variables showed weak correlations. First response time and resolution time had a correlation of approximately -0.01, indicating almost no linear relationship. Latitude and longitude had the strongest observed relationship at approximately -0.38. The very high correlation between Ticket ID and Created Month is not treated as a meaningful business relationship because Ticket ID is an identifier.

3.6 Statistical Summary

The average first response time was approximately 29.64 minutes and the average resolution time was approximately 24.31 minutes. The median resolution time was approximately 23.14 minutes. Agent interactions averaged 5.46, with values from 1 to 60. The average survey result was approximately 3.75 out of 5.

4. SLA Performance

First-response SLA analysis showed 2,019 tickets within SLA and 311 tickets with an SLA violation. Resolution SLA analysis showed 1,547 tickets within SLA and 783 tickets with an SLA violation. Therefore, resolution SLA violations were more numerous than first-response SLA violations in this dataset. This is a key operational finding because the project focuses on SLA compliance and resolution performance.

5. Visualization Findings

Ten visualizations were created using Matplotlib and Seaborn. They covered ticket status, priority, source, topic, first response-time distribution, resolution-time distribution, resolution time by priority, agent-group performance, monthly ticket volume, and numerical correlations.

No.	Visualization	Type	Purpose
1	Ticket Status Distribution	Bar	Compare ticket statuses
2	Ticket Priority Distribution	Bar	Compare priority levels
3	Ticket Source Distribution	Pie	Show source proportions
4	Ticket Topic Distribution	Bar	Identify common topics
5	First Response Time Distribution	Histogram	Examine response-time distribution
6	Resolution Time Distribution	Histogram	Examine resolution-time distribution
7	Resolution Time by Priority	Box plot	Compare distributions
8	Average Resolution Time by Agent Group	Bar	Compare support groups
9	Monthly Ticket Volume	Line	Identify monthly trends
10	Correlation Heatmap	Heat map	Visualize numerical relationships

6. Key Insights

Insight 1 — Resolution SLA requires attention

There were 783 resolution SLA violations compared with 311 first-response SLA violations. Resolution SLA therefore represents the larger category of SLA exception in this dataset.

Insight 2 — Email is the dominant support channel

Email accounted for 1,234 tickets, compared with 850 through Chat and 246 through Phone. Email therefore represents the largest share of support demand in the dataset.

Insight 3 — Product setup is the most common support topic

Product setup generated 630 tickets, the highest count among the listed topics. It represents a substantial part of the support workload in the dataset.

Insight 4 — Ticket volume changes across the year

Monthly ticket volume fluctuated rather than following a consistent upward or downward trend. January had approximately 224 tickets, while February had approximately 159. May rose to approximately 219 tickets.

Insight 5 — Resolution time varies across priority levels

Average resolution time was approximately 23.48 minutes for Low, 25.44 minutes for Medium, and 24.71 minutes for High priority. Medium priority had the highest average.

Insight 6 — Support groups have similar average resolution times

The 1st line support group averaged approximately 24.58 minutes and the 2nd line support group approximately 23.44 minutes. The difference is relatively small.

Insight 7 — Response time and resolution time show almost no linear correlation

The correlation between First response time and Resolution time was approximately -0.01, indicating almost no linear relationship between these two measures.

Insight 8 — Most tickets are completed

Closed and Resolved tickets together account for 1,912 of the 2,330 tickets. This means most records represent tickets that have reached a completed status.

7. Patterns, Correlations, and Anomalies

Email is consistently the largest support source, Product setup is the largest ticket topic, and resolution SLA violations are more numerous than first-response SLA violations. Resolution time is spread across a broad range, with many tickets resolved within shorter intervals. The correlation analysis did not identify a strong linear relationship between first response time and resolution time. The Ticket ID–Created Month correlation should not be interpreted as an operational relationship because Ticket ID is an identifier. The topic categories contain a capitalization inconsistency: “Pricing and licensing” and “Pricing and Licensing” appear separately. This is a data-standardization issue that could be addressed in a future version of preprocessing.

8. Overall Conclusion

The analysis provides an overall view of technical support operations in the dataset. The support workload is dominated by Email-based tickets, Low-priority tickets, and Product setup issues. Most tickets have reached a completed status, but resolution SLA violations are more numerous than first-response SLA violations. Average response and resolution times are both around 30 minutes or less, while resolution time differs moderately across priority levels and agent groups. The correlation analysis suggests that first response time alone does not have a meaningful linear relationship with total resolution time. Overall, the project demonstrates an end-to-end Python data analytics workflow covering cleaning, transformation, EDA, statistical analysis, visualization, insight generation, and recommendations.

9. Recommendations and Next Steps

1. Investigate the causes of resolution SLA violations and identify stages of the support process where delays occur.
2. Review the high volume of Product setup tickets to determine whether improved documentation, onboarding material, or self-service guidance could reduce repetitive support demand.
3. Review Email-based workflows because Email is the largest support channel in the dataset.
4. Compare agent-group performance using additional factors such as ticket priority, topic, complexity, and SLA outcome rather than average resolution time alone.
5. Standardize category names such as “Pricing and licensing” and “Pricing and Licensing” in future preprocessing.
6. Future analysis could examine SLA performance by month, source, topic, priority, and agent group to identify more specific operational drivers of SLA violations.

10. Final Project Summary

The project followed an end-to-end data analytics workflow: dataset loading and inspection, missing-value handling, duplicate checking, data-type correction, derived-column creation, filtering and aggregation, exploratory analysis, statistical summaries, correlation analysis, and ten visualizations. The findings were then converted into key insights, a conclusion, and practical next steps.